

Topic 16: Redox Equilibria

Students will be assessed on their ability to:

16.1	understand the terms 'oxidation' and 'reduction' in terms of electron transfer and changes in oxidation number, applied to <i>s</i> -, <i>p</i> - and <i>d</i> -block elements
16.2	know what is meant by the term 'standard electrode potential', $E^\ominus$
16.3	know that the standard electrode potential, $E^\ominus$, is measured in conditions of: i 298 K temperature ii 100 kPa pressure of gases iii 1.00 mol dm ⁻³ concentration of ions
16.4	know the features of the standard hydrogen electrode and understand why a reference electrode is necessary
16.5	understand that different methods are used to measure standard electrode potentials of: i metals or non-metals in contact with their ions in aqueous solution ii ions of the same element with different oxidation numbers
16.6	CORE PRACTICAL 12 Investigating some electrochemical cells.
16.7	be able to calculate a standard emf, $E^\ominus_{\text{cell}}$, by combining two standard electrode potentials
16.8	be able to write cell diagrams using the conventional representation of half-cells
16.9	understand the importance of the conditions when measuring an electrode potential, E
16.10	be able to use standard electrode potentials to predict the thermodynamic feasibility of a reaction
16.11	understand that $E^\ominus_{\text{cell}}$ is directly proportional to the total entropy change and to $\ln K$ for a reaction
16.12	understand the limitations of predictions made using standard electrode potentials, in terms of kinetic stability of systems and departure from standard conditions
16.13	know that standard electrode potentials are sometimes referred to as standard reduction potentials and can be listed as an electrochemical series
16.14	understand how standard electrode potentials can be used to predict the thermodynamic feasibility of disproportionation reactions
16.15	be able to carry out both structured and unstructured titration calculations involving redox reactions, including iron(II) ions and potassium manganate(VII) and sodium thiosulfate and iodine
16.16	be able to discuss the uncertainty of measurements and their implications for the validity of the final results

16.17	CORE PRACTICALS 13a and 13b Carry out redox titrations with both: i iron(II) ions and potassium manganate(VII) ii sodium thiosulfate and iodine
16.18	understand that fuel cells use the energy released on the reaction of a fuel with oxygen to generate a voltage <i>Knowledge that methanol and other hydrogen-rich fuels are used in fuel cells is expected.</i>
16.19	know the electrode reactions that occur in a hydrogen-oxygen fuel cell <i>Knowledge of hydrogen-oxygen fuel cells with both acidic and alkaline electrolyte is expected.</i>
	Further suggested practicals: i investigate the percentage of copper in brass, using iodine-thiosulfate titration ii investigate the percentage of iron in iron tablets, using potassium manganate(VII) titration iii prepare crystals of potassium iodate(VII) and measure their purity